

IB MATH AA SL

Lesson 15

Rational & Reciprocal Functions

Functions · 60-minute lesson

Syllabus SL 2.5 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. What is $1/x$ when x is very large?
2. What happens to $1/x$ as $x \rightarrow 0$?
3. Can $1/x$ ever equal 0?

Answers (click to reveal)

1. it gets close to 0
2. it grows huge ($\rightarrow \pm\infty$)
3. no — $y = 0$ is never reached

What we're learning today

LEARNING OBJECTIVES

- 1 Sketch**
the reciprocal function $y = 1/x$.
- 2 Identify**
vertical and horizontal asymptotes.
- 3 Analyse**
rational functions of the form $(ax+b)/(cx+d)$.
- 4 State**
the domain and range using asymptotes.

SUCCESS CRITERIA — I CAN...

- ✓
recognise the shape of $y = 1/x$
- ✓
find a vertical asymptote (denominator = 0)
- ✓
find a horizontal asymptote (ratio of leading coefficients)
- ✓
state domain and range from the asymptotes

The reciprocal function

$$y = 1/x$$

$$y = \frac{1}{x}$$

Vertical asymptote

$x = 0$ (can't divide by 0)

Horizontal asymptote

$y = 0$ (never reached)

Shape

two curves in opposite quadrants

ASYMPTOTE =

a line the curve approaches but never touches.

Vertical

where the function is undefined.

Horizontal

the value y heads to for large $|x|$.

Reciprocal asymptotes

QUESTION

State the asymptotes, domain and range of

$$y = 1/x.$$

SOLUTION

Denominator zero at $x = 0 \rightarrow$ vertical asymptote:

$$x = 0$$

For large $|x|$, $y \rightarrow 0 \rightarrow$ horizontal asymptote:

$$y = 0$$

Domain $x \neq 0$, range $y \neq 0$.

Your turn — reciprocals

ANSWER IN YOUR BOOK

1. Vertical asymptote of $y = 1/(x - 2)$?
2. Vertical asymptote of $y = 1/(x + 5)$?
3. Horizontal asymptote of $y = 1/x + 3$?

Answers (click to reveal)

1. $x = 2$
2. $x = -5$
3. $y = 3$

Rational functions

$(ax + b)/(cx + d)$

$$y = \frac{ax + b}{cx + d}$$

Vertical asymptote

set the denominator = 0

Horizontal asymptote

$y = a/c$ (ratio of the x-coefficients)

METHOD

1.
denominator = 0 → vertical asymptote
2.
leading coefficients → horizontal asymptote
3.
domain & range exclude those values.

Asymptotes of a rational

QUESTION

Find the asymptotes of

$$y = (2x + 1)/(x - 3).$$

SOLUTION

Vertical: denominator zero $\rightarrow x - 3 = 0$:

$$x = 3$$

Horizontal: ratio of x-coefficients 2/1:

$$y = 2$$

Domain $x \neq 3$, range $y \neq 2$.

Exam-style question

Consider $f(x) = (3x - 2)/(x + 1)$.

- (a) Write down the equation of the vertical asymptote. [1]
- (b) Write down the equation of the horizontal asymptote. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $x = -1$ (A1)
- (b) $y = 3$ (ratio 3/1) (A1)(A1)

Common mistakes



Vertical asymptote

Set the DENOMINATOR to zero, not the numerator.



Horizontal asymptote

Use the ratio of the leading coefficients, a/c .



Touching

A curve never crosses a vertical asymptote.



Domain

Exclude the x that makes the denominator zero.

Mixed practice

QUESTIONS

1. Vertical asymptote of $y = 1/(x - 6)$?
2. Asymptotes of $y = (x + 4)/(x - 2)$?
3. Horizontal asymptote of $y = (5x)/(x + 1)$?
4. Domain of $y = 1/(x + 3)$?

Answers (click to reveal)

1. $x = 6$
2. $x = 2$ and $y = 1$
3. $y = 5$
4. $x \neq -3$

Plenary & exit ticket

KEY FORMULAS

$$y = \frac{1}{x} : x = 0, y = 0$$

$$y = \frac{ax + b}{cx + d} : y = \frac{a}{c}$$

EXIT TICKET — before you leave

1. Vertical asymptote of $y = 1/(x - 1)$?
2. Horizontal asymptote of $y = (2x+3)/(x-1)$?
3. Domain of $y = 1/x$?

Answers (click to reveal)

1. $x = 1$
2. $y = 2$
3. $x \neq 0$

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Rational & reciprocal functions.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 17

Trigonometry begins — radians, arcs and sectors.

$$\pi \text{ rad} = 180^\circ$$